

HAFSA FATHIMA

Artificial Intelligence & Machine Learning Undergraduate
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Education

Jayaprakash Narayana College of Engineering, Hyderabad

Expected May 2027

Bachelor of Technology (B.Tech) in Artificial Intelligence & Machine Learning

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Natural Language Processing, Deep Learning

Technical Skills

- **Programming Languages:** Python, Java, JavaScript, SQL, C
- **AI / Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, MediaPipe
- **LLM & Agentic AI:** LangChain, LangGraph, RAG Architecture, Multi-Agent Systems, Semantic Retrieval, Prompt Engineering
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Feature Engineering, Power BI
- **Web Technologies:** FastAPI, Node.js, REST APIs, HTML5, CSS3, JavaScript (ES6+)
- **Tools & Platforms:** Git, GitHub, Linux CLI, VS Code, Google Colab, PyCharm

Selected Projects

- **RefluxAI-Advanced – Multi-Agent AI Code Generation and Review Platform** GitHub
Autonomous multi-agent software engineering platform that generates, reviews, tests, secures, documents, and deploys code from natural-language requirements.
 - Engineered a 6-agent workflow using LangGraph and Groq LLMs for code generation, review, testing, performance optimization, security auditing, and documentation.
 - Integrated GitHub Pull Request automation and quality evaluation metrics including complexity analysis, test coverage assessment, and vulnerability detection.
- **AI Log Analyzer – LLM Powered Automated Debugging System** GitHub
LLM-powered debugging assistant that analyzes application logs, identifies root causes, and generates structured troubleshooting recommendations.
 - Developed an automated log analysis pipeline that parses system logs, detects failures, and routes errors through an LLM-powered diagnostic workflow.
 - Automated error categorization and root-cause identification to streamline debugging and software maintenance workflows.
- **AskDuo – Retrieval-Augmented AI Knowledge Assistant** GitHub
RAG-powered enterprise knowledge assistant that retrieves information from internal documentation and generates source-grounded responses.
 - Built a modular Retrieval-Augmented Generation pipeline using LangChain, ChromaDB, OpenAI Embeddings, and GPT-4o-mini for semantic search across organizational knowledge bases.
 - Implemented persistent vector storage, semantic retrieval, and citation-based answer generation to improve retrieval accuracy and reduce hallucinations.
- **Wall-Climbing Window Cleaner Robot** Mar 2026
Autonomous wall-climbing robotic system designed for vertical surface navigation and automated window-cleaning operations.
 - Designed and developed a mobile-controlled robotic prototype integrating ESP32, DC geared motors, motor driver modules, buck converter, BLDC motor, and wireless communication.
 - Implemented wall-adhesion and navigation mechanisms for stable movement across vertical surfaces while carrying cleaning equipment.

Achievements

- Co-authored and published 2 machine learning research papers in academic journals.
- Active **LeetCode** practitioner, focusing on arrays, trees, graphs, recursion, and dynamic programming.
- Presented AI/ML projects at **Science Fest 2025** and **ESCI Hyderabad Tech Expo**.
- Built and showcased AI solutions in hackathons and technical competitions.

Research & Publications

- “Car Price Prediction Using Machine Learning”, published in *Research and Reviews: Advancement in Cyber Security* (2025).
- “Future Proofing Real Estate: Machine Learning for Price Predictions”, published in *Advancement of Computer Technology and Its Applications* (2025).

Hackathons & Competitions

- Built **LegalShe**, an AI-powered multilingual legal assistance platform during the **DevQueens Women's Hackathon**.
- Participated in **CBIT Hackathon** and **Specathon**, developing AI-based solutions under time-constrained environments.

Leadership & Activities

- **English Anchor – College-Level Technical Event:** Hosted and moderated technical sessions, coordinated speaker transitions, facilitated audience engagement, and collaborated with organizing teams to ensure successful execution of college-level technical events.